💡 **Useful Formulas for kinematics**

(1) $v = v_0 + at$

(2) $\Delta x = v_0 t + \frac{1}{2}at^2$

(3) $v^2 = {v_0}^2 + 2a\Delta x$

Q1: A bicycle's velocity increases steadily from $4\,\mathrm{m/s}$ to $16\mathrm{m/s}$ in $6\,\mathrm{s}$.

(a) What is its average acceleration?

(b) Draw the Velocity-Time (v-t) graph for this bicycle. Calculate the displacement from $t = 0$ to $6$ $\mathrm{s}$ from v-t graph.

Q2: A car accelerates from rest at a constant rate of $3\,\mathrm{m/s}^2$.

(a) How much time does it takes for the car to reach a final velocity of $15\,\mathrm{m/s}$?

(b) Calculate the distance the car traveled during this time.

Q3: A train is traveling at $30\,\mathrm{m/s}$. The engineer hits the brakes, and the train comes to a complete stop ($v = 0\,\mathrm{m/s}$) over a distance of $90\,\mathrm{m}$.

(a) Calculate the train's constant acceleration.

(b) How many seconds it took for the train to stop.

Q4: A toy rocket is launched straight up into the air. Gravity acts on it with $a = -10\,\mathrm{m/s}^2$ . It reaches a maximum height of $20\,\mathrm{m}$ before falling back down. At its maximum height, its instantaneous velocity is exactly $0\,\mathrm{m/s}$.

(a) Calculate its initial launch velocity.

(b) Draw the Velocity-Time (v-t) graph for the rocket from launch until it reaches its highest point. Use the graph to find out how many seconds it took to reach the top.

Q5: A cyclist rides in a two-stage motion.
(a) Stage 1: She starts from rest and accelerates at $3\,\mathrm{m/s}^2$
for $4\,\mathrm{s}$. What is her speed at the 4-second mark? How far did she ride during Stage 1?
(b) Stage 2: She then maintains that top speed (constant velocity) for another $5\,\mathrm{s}$. How far did she ride during Stage 2?
(c) Draw a v-t graph for this entire 9-second ride and calculate the total displacement via the graph.

Q6: Car A is driving at $30\,\mathrm{m/s}$. Car B is $75\,\mathrm{m}$ directly ahead of Car A, driving in the same direction at a constant velocity of $10\,\mathrm{m/s}$. The driver of Car A hits the brakes, creating a constant deceleration of $-2\,\mathrm{m/s}^2$.
(a) The danger of a collision only exists while Car A is moving faster than Car B. Calculate how many seconds it takes for Car A to slow down matching Car B's speed.
(b) Using the time you found in (a), calculate the displacement of Car A and the displacement of Car B during this specific time period.
(c) Assume Car A started at position $x_A = 0\,\mathrm{m}$. This means Car B started at $x_B = 80\,\mathrm{m}$. What are their final positions at time t? Based on these positions, did Car A crash into Car B?
(d) When they exactly crush to each other, which physical quantity of two cars will be the same? List the equation and solve for the exact second the crash happens.